

Parker Lovill

ID#: 00127715

plovill@my.athens.edu

CS514

Question 1:

type 'a rose = Node of 'a * 'a rose list

A rose tree is a node holding a value adding a list of child subtrees.

size : 'a rose -> int

Counts the total number of nodes, 1 for the current node, plus the sizes of all children summed up.

map : ('a -> 'b) -> 'a rose -> 'b rose

Applies f to every value in the tree, holding the tree's shape.

fold : ('a -> 'b list -> 'b) -> 'a rose -> 'b

This lets you reduce a whole tree down to a single value of any type b, given a combining function.

Question 2:

C++ Identifiers [a-zA-Z_][a-zA-Z0-9_]*

US Phone # \(\d{3}\) \d{3}-\d{4}

Floating Point # [+~]? \d+(\.\d+)?

Binary Palindromes. 000 | 010 | 101 | 111

(3+(2*6)/2)

Factor

"("

Expr

Expr

Term

Factor

number: 3

"+"

Term

Term

Factor

"("

Expr

Term

Term

Factor

number: 2

"*"

Factor

number: 6

")"

"/"

Factor

number: 2

4*(3+2)*4

Expr

Term

Term

Term

Factor

number: 4

"*"

Factor

"("

Expr

Expr

Term

Factor

number: 3

"+"

Term

Factor

number: 2

")"

"*"

Factor

number: 4

42*1+3*(80+90)

Expr

Expr

Term

Term

Factor

number: 42

"*"

Factor

number: 1

"+"

Term

Term

Factor

number: 3

"*"

Factor

"("

Expr

Expr

Term

Factor

number: 80

"+"

Term

Factor

number: 90

)"

Question 3:

Expr $\rightarrow$ Expr + Term

| Expr - Term

| Term

Term $\rightarrow$ Term * Factor

| Term / Factor

| Factor

Factor $\rightarrow$ + Factor

| - Factor

| (Expr)

| number

(3 + (-3)) * 4

Expr

Term

Term

Unary

Factor

"("

Expr

Expr

Term

Unary

Factor

number: 3

"+"

Term

Unary

"_"

Unary

Factor

number: 3

")"

"*"

Unary

Factor

number: 4

Question 4:

```
[→ Assignment01 git:(main) ✗ dune exec ./bin/main.exe -- "(3+-3)*4"

Input:
(3+-3)*4

AST:
Mul
  Add
    Int(3)
    Neg
      Int(3)
  Int(4)

Fully parenthesized:
((3 + (-3)) * 4)

Value:
0
[→ Assignment01 git:(main) ✗ grep "Minus" bin/token.ml
| Minus
| Minus -> "Minus"
```

```
→ Assignment01 git:(main) ✗ dune exec ./bin/main.exe -- "4*(3+2)*4"
```

```
Input:  
4*(3+2)*4
```

```
AST:  
Mul  
  Mul  
    Int(4)  
    Add  
      Int(3)  
      Int(2)  
  Int(4)
```

```
Fully parenthesized:  
((4 * (3 + 2)) * 4)
```

```
Value:  
80
```

```
→ Assignment01 git:(main) ✗ dune exec ./bin/main.exe -- "42*1+3*(80+90)"
```

```
Input:  
42*1+3*(80+90)
```

```
AST:  
Add  
  Mul  
    Int(42)  
    Int(1)  
  Mul  
    Int(3)  
    Add  
      Int(80)  
      Int(90)
```

```
Fully parenthesized:  
((42 * 1) + (3 * (80 + 90)))
```

```
Value:  
552
```

```
→ Assignment01 git:(main) ✗ dune exec ./bin/main.exe -- "1 - 2 - 3"
```

```
Input:  
1 - 2 - 3
```

```
AST:  
Sub  
  Sub  
    Int(1)  
    Int(2)  
    Int(3)
```

```
Fully parenthesized:  
((1 - 2) - 3)
```

```
Value:  
-4
```

```
[→ Assignment01 git:(main) ✗ dune exec ./bin/main.exe -- "--3"
```

```
Input:  
--3
```

```
AST:  
Neg  
  Neg  
    Int(3)
```

```
Fully parenthesized:  
(-(-3))
```

```
Value:  
3
```

```
→ Assignment01 git:(main) ✗
```